STAT 361 Assignment 3 Question 3

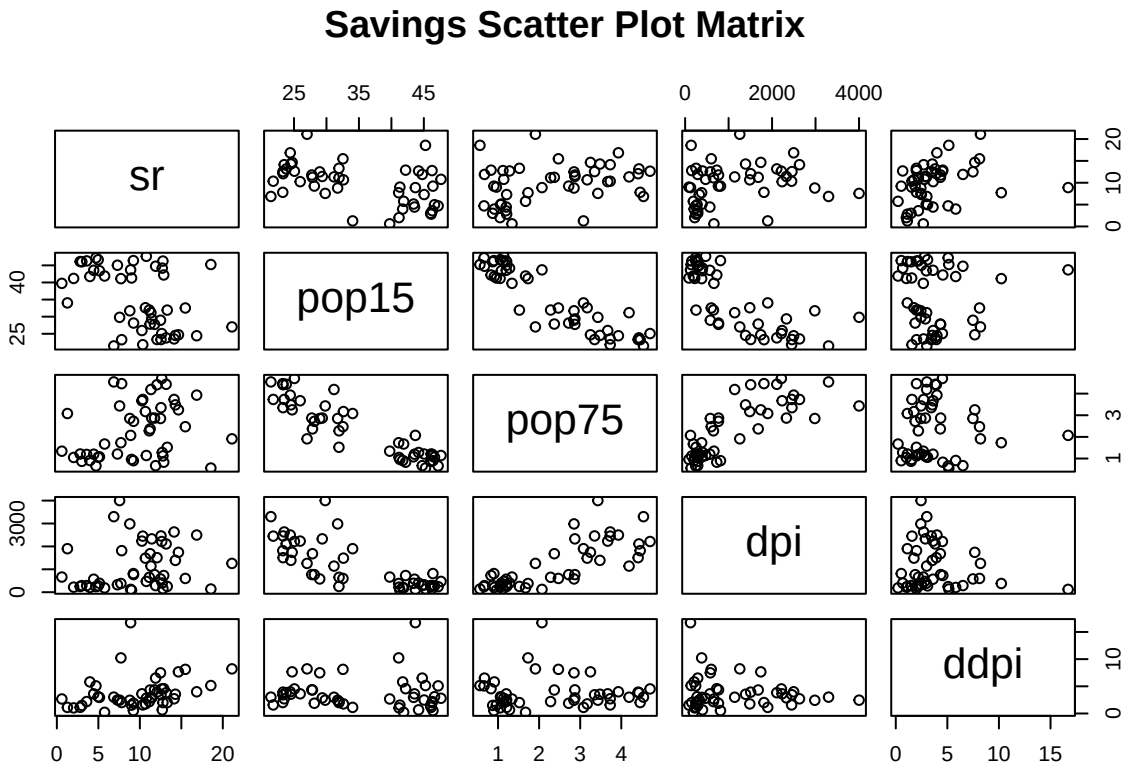
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```
savings <- read.table("Savings.txt", header = TRUE)
```

a) Scatter Plot Matrix

```
pairs(savings, main = "Savings Scatter Plot Matrix")
```



Looks like `ddpi` is not entirely useful, it is constant on one axis for every other variable. `pop75` and `dpi` seem to have a steady, positively correlated, linear trend. Meanwhile `pop15` and `pop75` are negatively correlated and linearly-trended, however the axes have very different medians. Otherwise, nothing else really comes to mind. Perhaps some slight bimodality between `pop15` and `sr`?

b) SLR for dpi ~ pop15

```
slr <- lm(dpi ~ pop15, data = savings)
summary(slr)
```

```
##
## Call:
## lm(formula = dpi ~ pop15, data = savings)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1142.65  -355.24     4.95   213.58  2429.48
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4098.51     378.87  10.818 3.15e-14 ***
## pop15         -84.74      10.48  -8.088 2.19e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 651.3 on 46 degrees of freedom
## Multiple R-squared:  0.5872, Adjusted R-squared:  0.5782
## F-statistic: 65.42 on 1 and 46 DF,  p-value: 2.192e-10
```

This model can be described as:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_{pop15}$$

for beta_1 being the coefficient for the covariate `pop15`, beta_0 being the intercept/bias, x_pop15 being the covariate `pop15`, and y_hat being the predicted value of `dpi`.

There is significant association between the variables `pop15` and `dpi`, which is not entirely expected honestly but having a second look at the scatterplot I can see how that would happen. Technically it does follow a linear trend, but the bimodality threw me off. It does seem a bit too clustered at the higher `pop15` values and wide at the earlier values, which also threw me off. Overall, we should be cautious as to the efficacy of the model because the lower R^2 scores of ~0.59 are not too high, and we have about 40% unexplained variance, which is significant.

c) Sample Correlation Coefficient

```
# Equivalent to the formula in the class slides
samp_corr = cor(savings$pop15, savings$dpi, method = "pearson")
samp_corr^2
```

```
## [1] 0.5871516
```

This sample correlation coefficient squared is exactly the calculated R^2 value to the `slr` model fit in b).

d) Multliple Linear Regression Model

```r
mlr <- lm(sr ~ ., data = savings)
summary(mlr)
```

```
##
## Call:
## lm(formula = sr ~ ., data = savings)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.251 -2.652 -0.218  2.387  9.755
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 29.6809865  7.5684194   3.922 0.000312 ***
## pop15       -0.4865933  0.1495275  -3.254 0.002218 **
## pop75       -1.7107905  1.1009475  -1.554 0.127532
## dpi         -0.0004923  0.0009608  -0.512 0.610992
## ddpi         0.4218624  0.1994379   2.115 0.040240 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.856 on 43 degrees of freedom
## Multiple R-squared:  0.3467, Adjusted R-squared:  0.286
## F-statistic: 5.706 on 4 and 43 DF,  p-value: 0.000894
```

This model can be described as:

$$\hat{Y} = \hat{\beta}X$$

for the vector beta with elements beta_0, ... beta_4, for the matrix X with vectors 1, x_1, ..., x_4 and the predicted values of Y_hat being `sr`. x_1 is `pop15`, x_2 is `pop75`, x_3 is `dpi`, and x_4 is `ddpi`. Each beta value index corresponds to the respective x vector index, and beta_0 corresponds to the vector of 1s.

The fit of the model seems quite poor, with a very low R^2 value of ~0.34 (and and even lower adjusted R^2 value), two highly redundant values of `pop75` and `dpi` that were not statistically significant to the model, thereby hurting its performance. Somehow the F-test parameter has a low p-value, meaning the test of all slopes would be passed at an alpha = 0.05 level, but the other metrics would make me uncomfortable to accept this model.

The estimated coefficient beta_4, corresponding to `ddpi`, means that for every percentage increase in the rate-of-change of disposable income (`dpi`), we increase the predicted ratio of savings (`sr`) by around 0.42 units.

e) ANOVA and MLR variance

```r
anova(mlr)
```

```
## Analysis of Variance Table
##
## Response: sr
##           Df Sum Sq Mean Sq F value    Pr(>F)
## pop15      1 201.13 201.130 13.5271 0.0006495 ***
```

```
## pop75      1  55.48  55.484  3.7316 0.0599986 .
## dpi        1  16.23  16.230  1.0916 0.3019614
## ddpi       1  66.53  66.527  4.4743 0.0402405 *
## Residuals 43 639.35  14.869
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note the Sum Sq Residuals (SS(Res)) value of 639.35.

```
mlr_sigma = sigma(mlr)
mlr_sigma
```

```
## [1] 3.855995
```

```
# 43 degrees of freedom in the residuals according to the mlr model
# Calculated using formula in class
ms_res = mlr_sigma^2 * 43
ms_res
```

```
## [1] 639.3541
```

We see that the fitted sigma_hat value of the MLR model, when squared and multiplied by the degrees of freedom of the residuals, is exactly the mean-squared residuals. This means that we can efficiently use the mean-squared residuals to estimate the variance of an MLR model.